

PRATEEK DWIVEDI

प्रतीक द्विवेदी

KD213, Indian Institute of Technology Kanpur

pdwivedi@cse.iitk.ac.in ◇ prateekdwivedi.in

RESEARCH INTERESTS

Broadly I am interested in the Theoretical Aspects of Computer Science. Specifically, my interest lies in Algebraic and Computational Complexity Theory.

EDUCATION

Indian Institute of Technology Kanpur

Doctor of Philosophy

Computer Science & Engineering

Advisor: **Prof Nitin Saxena**

Kanpur, India

2018 - Ongoing

KCC Institute of Technology and Management

Bachelor of Technologies

Computer Science & Engineering

Gautam Buddha Nagar, India

2013 - 2017

PUBLICATIONS

Lower Bounds for the Sum of Small-Size Algebraic Branching Programs.

with C.S. Bhargav and Nitin Saxena

Under review.

2023

Demystifying the border of depth-3 algebraic circuits.

with Pranjal Dutta and Nitin Saxena.

62nd Annual Symposium on Foundations of Computer Science (FOCS)

Invited in the special issue of SIAM Journal on Computing (SICOMP)

2021

Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits

with Pranjal Dutta and Nitin Saxena.

36th Computational Complexity Conference (CCC)

The full version is currently under review in Theory of Computing (ToC)

2021

ACADEMIC TALKS AND SEMINARS

Treading the Border Complexity and Identity Testing Paradigms.

PhD State of The Art Seminar, IIT Kanpur

2022

Deterministic identity testing paradigms for bounded top-fanin depth 4 circuits.

Invited Talk, CS Theory Seminar at Georgetown University.

2021

Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits.

- *Workshop on Algebra and Computation (WAC), Göteborg*

2023

- *7th Workshop on Algebraic Complexity Theory (WACT), Warwick*

2023

- *CS Theory Seminar at Georgetown University.*

2021

• <i>Conference Presentation, (CCC)</i>	2021
Information-Theoretic And Algorithmic Thresholds For Group Testing <i>PhD Comprehensive Evaluation, IIT Kanpur.</i>	2020
On Approximative Closure of Algebraic Complexity Classes <i>SIGTACS, IIT Kanpur.</i>	2019

CONFERENCES AND WORKSHOP ATTENDED

Summer of Counting and Algebraic Complexity <i>IT University of Copenhagen (ITU) in Copenhagen, Denmark</i>	2023
Workshop on Algebra and Computation (WAC) <i>Chalmers University of Technology, Göteborg</i>	2023
7th Workshop on Algebraic Complexity Theory (WACT) <i>The University of Warwick, UK</i>	2023
2nd Swiss Winter School on Theoretical Computer Science <i>Jointly organized by EPFL and ETH Zurich in Zinal</i>	2023
62nd Annual Symposium on Foundations of Computer Science (FOCS) <i>Denver, Colorado - Attended Virtually</i>	2022
36th Computational Complexity Conference (CCC), 2021. <i>Toronto, Ontario, Canada - Attended Virtually</i>	2021
48th International Colloquium on Automata, Languages, and Programming (ICALP) <i>Glasgow, Scotland - Attended Virtually</i>	2021
52nd ACM Symposium on Theory of Computing (STOC) <i>Chicago, US - Attended Virtually</i>	2020
Workshop on Algebraic Complexity Theory <i>ICTS, Bengaluru</i>	2019

ACADEMIC ACHIEVEMENTS

Invited to participate in Summer of Counting and Algebraic Complexity at ITU Copenhagen <i>Full financial support.</i>	2023
Selected to attend Swiss Winter School on Theoretical Computer Science <i>Partial financial support.</i>	2023
Recipient of financial support from Microsoft Research India and IARCS-ACM India. <i>To attend Academic conference.</i>	2022
Ranked 23 in Joint Entrance Screening Test(JEST).	2018
Ranked 1179 in Graduate Aptitude Test in Engineering (GATE). <i>Among 1,00,000 candidates.</i>	2017
Presented Undergraduate project at University level Science and Technology Conference.	2017
Academic Excellence Award. <i>KCC ITM, Greater Noida</i>	2014

TEACHING EXPERIENCE

Tutor at CSE, IIT Kanpur

- Introduction to Computing 2022
- Data Structures and Algorithms 2019

Teaching Assistant at CSE, IIT Kanpur

- Computational Complexity 2021
- Modern Cryptology 2021
- Mathematics for Computer Science 2020
- Data Structures and Algorithms 2018
- Introduction to Computing 2018

Teaching Assistant in Massive Online Courses

- Introduction to Cryptography | eMasters, IIT Kanpur 2022
- Randomized methods in Computational Complexity | NPTEL 2021
- Arithmetic Circuit Complexity | NPTEL 2020
- Modern Algebra | NPTEL 2019

PROFESSIONAL CONTRIBUTIONS

Reviewed research papers for conferences and journals

STOC, FOCS, CCC, ITCS, SODA, SIAM Journal on Computing.

Regularly organise talks and seminars in a special interest group

SIGTACS. CSE, IIT Kanpur.

Co-offered a short course on Mathematics for Computer Science.

Adhyayan 2023. Summer school, CSE, IITK

PERSONAL DETAILS

Name	Prateek Dwivedi (प्रतीक द्विवेदी)
Gender and Pronouns	Male and He/Him.
Permanent Address	A106, Bulland Heights, Crossings Republik Ghaziabad (U.P) - 201016, India
Date and place of birth	21 st Feb 1996, Lucknow, India
Nationality	Indian (भारतीय)
Contact	+91 8800 291 032 +91 512 259 7123

REFERENCES

Nitin Saxena

Professor

IIT Kanpur

nitin@cse.iitk.ac.in

Nikhil Balaji

Assistant Professor

IIT Delhi

nbalaji@cse.iitd.ac.in